



**FACULTY OF TECHNOLOGY AND ENGINEERING
DEVANG PATEL INSTITUTE OF ADVANCE TECHNOLOGY
AND RESEARCH**

DEPARTMENT OF INFORMATION TECHNOLOGY

A.Y 2026 – 27 Semester – 3

LAB MANUAL

CSUC201 Fundamentals of Data Structures and Algorithms



PRACTICAL 1: ARRAY OPERATIONS - TRAVERSAL, ROTATION AND FREQUENCY ANALYSIS

PROBLEM-1

DIFFICULTIES:

- I was confused about how to move elements from the right side to the left side.
- The value of h was very large, so I was not sure how large it could be.
- I was not clear about the concept of array rotation.

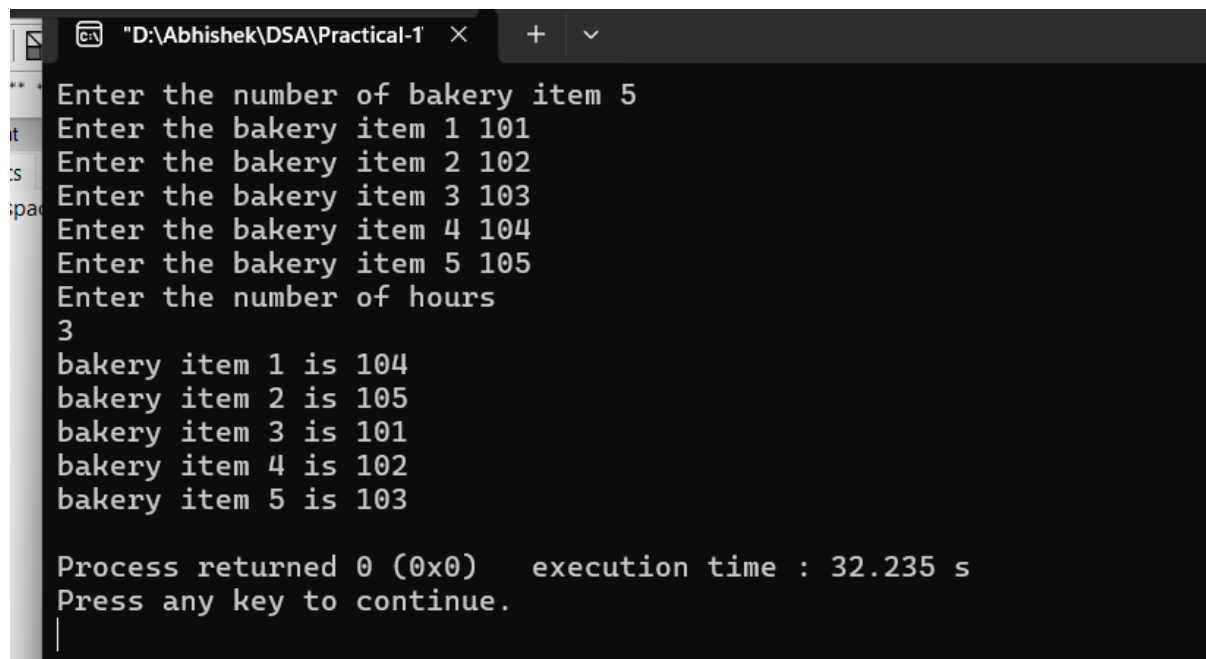
CODE:

<https://github.com/abhishekipatadiya28-hash/DATA-STRUCTURE-ANDALGORITHM/tree/main/Practical-1>

ERRORS:

- I used the wrong loop for rotation.
- I forgot to handle the first element properly after rotation.

OUTPUT:

A screenshot of a C++ program execution in a Windows command prompt. The window title is "D:\Abhishek\DSA\Practical-1". The program prompts the user to enter the number of bakery items (5), then enters each item's ID (101, 102, 103, 104, 105). It then prompts for the number of hours to rotate (3). The output shows the items after rotation: item 1 is 104, item 2 is 105, item 3 is 101, item 4 is 102, and item 5 is 103. The process returned 0 (0x0) and took 32.235 seconds to execute.

```
"D:\Abhishek\DSA\Practical-1" x + v
Enter the number of bakery item 5
Enter the bakery item 1 101
Enter the bakery item 2 102
Enter the bakery item 3 103
Enter the bakery item 4 104
Enter the bakery item 5 105
Enter the number of hours
3
bakery item 1 is 104
bakery item 2 is 105
bakery item 3 is 101
bakery item 4 is 102
bakery item 5 is 103

Process returned 0 (0x0)   execution time : 32.235 s
Press any key to continue.
```

SOLUTION:

First, I understood the problem and learned that I had to use array rotation. I also took help from my faculty to understand the concept. After that, I was able to find the correct solution.

OUTCOMES:

- Learned how to perform left rotation in an array.
- Understood the concept of array rotation.
- Learned that a very large value of h can affect program performance.

PROBLEM-2

DIFFICULTIES:

- The problem had a large amount of data, around 100,000 records, so I had to write the code carefully.
- I had to understand that if one book was borrowed by 4 students, it should be counted as only 1 borrowed book.

ERRORS:

- I used the wrong loop condition ($i \leq n$).
- I made mistakes while using the array and comparing values.

SOLUTION:

First, I checked the errors in my code. I changed the loop condition to ($i < n$) and tested it. I also learned about array indexing and understood when to use `arr[i+1]`.

OUTPUT:

```
Enter the number of Books5
Enter the Book ID 1
102
Enter the Book ID 2
110
Enter the Book ID 3
320
Enter the Book ID 4
558
Enter the Book ID 5
20
how many time book number 1 is borrowed
2
how many time book number 2 is borrowed
1
how many time book number 3 is borrowed
4
how many time book number 4 is borrowed
0
how many time book number 5 is borrowed
3
The book which is borrowed more than once are 102
The book which is borrowed more than once are 320
The book which is borrowed more than once are 20

Process returned 0 (0x0)   execution time : 30.183 s
Press any key to continue.
```

OUTCOMES:

- Understood the difference between $O(n)$ and $O(n^2)$ complexity.
- Learned that nested loops take $O(n^2)$ time.
- Learned that a frequency table takes approximately $O(n)$ average time.
- Understood that the frequency table is better for large data.

PROBLEM-3**DIFFICULTIES:**

- I was confused about whether I had to find the longest word or the longest sentence.
- I was confused about which word to select when two or more words had the same length.

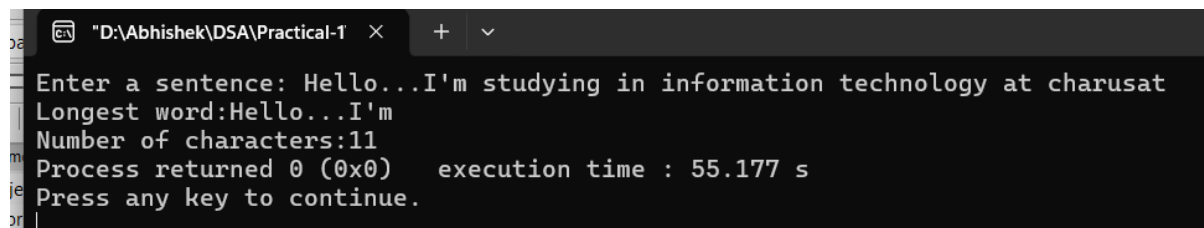
ERROR:

- I used the wrong loop condition.
- I compared the words instead of comparing their lengths.

SOLUTION:

I took the complete sentence as input and checked it character by character. I separated each word when a space was found and compared its length with the current longest word.

If the new word was longer, I stored it as the longest word. I continued this process until the sentence ended. Finally, I checked the last word and printed the longest word with its length.

OUTPUT:

```
"D:\Abhishek\DSA\Practical-1" X + v
Enter a sentence: Hello...I'm studying in information technology at charusat
Longest word:Hello...I'm
Number of characters:11
Process returned 0 (0x0)    execution time : 55.177 s
Press any key to continue.
```

OUTCOMES:

- Learned how to separate a sentence into individual words.
- Learned how to compare word lengths.
- Understood how to handle the last word.
- Learned how to handle words with equal lengths.
- Understood the difference between $>$ and $>=$.

KEY QUESTIONS:-

1. [Problem 1] The display order repeats itself after a certain number of hours regardless of how large h gets. Without trying any specific example, explain why this repetition is inevitable and what determines when it first repeats.

ANSWER:-

The repetition is inevitable because a display has a fixed number of possible states. After a certain number of hours, the display must return to a state that it has already shown. The first repetition depends on the cycle of the display system. For example, if the display

represents a 12-hour clock, the same display repeats every 12 hours because there are only 12 different hour positions. So, no matter how large h becomes, the display can be reduced to its repeating cycle using the modulo (%) operation. The repetition is inevitable because the display has a finite number of possible states. The first repetition occurs after completing one full cycle, which is determined by the number of states in that cycle.

2. [Problem 2] If the same book ID appears three times in the log, should it appear once or three times in your output? Does your solution handle this correctly, and how would you verify it?

ANSWER:-

If the same book ID appears three times, the answer depends on what the problem asks. If the problem is asking for distinct/unique book IDs, then it should appear only once. If the problem asks for unique book IDs, an ID appearing three times should appear only once. I would verify this by testing the program with duplicate IDs and checking the output.

3. [Problem 3] The editor later decides that punctuation attached to a word (e.g. "hello," or "great!") should not count toward its length. How would this change your solution, and at which step would you handle it?

ANSWER:-

I would remove punctuation from the word before calculating its length. This change would be handled during the input/preprocessing step, before finding the word length. For example, hello, would be treated as hello, so its length would be 5.